

Xiang Yang (杨翔)

Ph.D. Candidate in Human Factors Engineering & Ergonomics | Virginia Tech, USA | Multimodal & Immersive Interaction (Haptics · VR · Wearable Sensing)

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GitHub: [Personal](#) and [Work](#) · Languages: Mandarin (native), English (full professional proficiency)

SUMMARY

Product-value-oriented human factors scientist (Ph.D.) with hands-on engineering delivery, specializing in wearable smart hardware and multimodal interaction experience (visual, auditory, and haptic), with an interdisciplinary background spanning mechanical engineering and cognitive psychology. Core strength: closing the full loop from needs discovery, experimental design, and prototype development to data modeling and deployable standards. Builds multimodal sensor systems centered on EMG and motion capture, using both inertial (IMU) and optical tracking, to turn vague subjective experience into objective, measurable engineering metrics such as cognitive load and fatigue limits, and is well grounded in complementary methods including EEG and eye tracking. Full-stack prototyping (embedded and mobile) plus data analysis enable rapid concept validation and identification of the key drivers of experience. Independently built and deployed a personal AI system (Digital Me) that applies human-factors principles such as managing attention, revealing information gradually, and keeping the person in control. Now focused on bringing this toolkit to immersive entertainment: quantifying immersion and presence, and developing multimodal HCI technologies from concept to implementation for location-based entertainment (LBE) experiences.

CORE SKILLS

Human Factors Research & Experience Quantification

- **Multimodal experience quantification:** build sensor systems that capture EMG and motion-capture data, using both inertial (IMU) and optical (Vicon) tracking, and combine them with NASA-TLX and psychophysical methods so subjective feedback is paired with objective measurement, turning fuzzy experience into quantifiable parameters such as cognitive load. Familiar with EEG and eye-tracking methods and how they are used in experience research.
- **Individual calibration & adaptation:** rather than one-size-fits-all thresholds, calibrate and model interaction to individual perceptual differences, for example defining a metric as the deviation from a person's own baseline. This is a central theme of my dissertation and of my published work on adapting vibrotactile warning systems to individual users.
- **Experimental design (DOE):** expert in multi-factor experimental design; develop scientifically rigorous human-factors study protocols for different product forms (wearable, handheld, desktop, and immersive VR), ensuring data reliability and validity.

Smart Hardware & Interaction Prototyping

- **Integrated hardware and software development:** independently build high-fidelity interactive prototypes, spanning embedded firmware (Arduino / ESP32 / C++) with IMU and other sensor integration, to mobile apps (Flutter / iOS / Android), for rapid concept validation.
- **Multimodal feedback design (Visual / Audio / Haptics):** coordinated visual-auditory-haptic design; expert in vibration waveform tuning, frequency response, and timing optimization to raise the immersion and information-transfer efficiency of wearable devices.
- **Structural integration & engineering optimization:** proficient in SolidWorks / Pro-E precision structural design with 3D-printing iteration, delivering prototypes that meet early engineering validation (EVT) needs. CAD skills transfer directly to Fusion 360.
- **Interactive development toolchain:** Python (advanced) · Unity (working knowledge) · TouchDesigner (basic; node-based real-time visual pipelines).

Programming & Data Analysis

- **Complex data mining:** proficient in Python / R / MATLAB / JMP for large-scale human-factors data cleaning and statistical modeling; skilled at isolating the key drivers of user experience from noisy multi-source data to support product decisions.
- **Building and evaluating AI tools:** built AI-assisted workflows that analyze large volumes of user feedback and multi-language sources, and set up side-by-side comparisons to judge objectively which tools actually perform best; comfortable across Python, web APIs, and cloud deployment.

EDUCATION

Ph.D. | Industrial & Systems Engineering: Human Factors Engineering & Ergonomics (GPA 3.91/4.00)

2021 – 2026

Virginia Tech, USA, program ranked #7 in the U.S. (U.S. News, Industrial & Systems Engineering)

Core coursework: Human Factors Research Design (A) · System Safety Analysis (A) · Advanced Movement Assessment (A) · Statistical Research Methods (A)

Dissertation: Investigating Individual Differences, Signal Design, and Adaptation in Vibrotactile Signal Mapping. Studies how people interpret wearable vibration (haptic) warning signals in immersive VR and on-body devices, and how signal design and short training reshape their responses. Closely aligned with multimodal and immersive human-computer interaction.

M.S. | Mechanical Engineering, University of Arizona, USA 2019

B.S. | Mechanical Engineering, Lanzhou University of Technology 2015

PROJECT EXPERIENCE

Lead Researcher, Human Factors & Interaction | Intelligent Wearable Safety System, Virginia Tech 2021 – Present

- **Project:** for high-risk, dynamic environments, lead R&D of a distributed wearable safety system built on multimodal interaction (haptic, auditory, and visual); coordinated head-and-torso haptic feedback delivers complex information when vision is limited. This work is the basis of my dissertation on how people interpret directional haptic signals.
- **Prototyping:** independently delivered full-stack hardware and software across multiple devices, building distributed haptic arrays covering head and torso; used 3D printing and flexible-electronics integration to solve sensor fit, heat dissipation, and ergonomic wearing comfort across rigid helmets and fabric vests.
- **Quantifying UX:** designed immersive, high-workload VR experiments; used a Vicon motion-capture system with NASA-TLX to quantify perception thresholds and reaction latency for dynamic vibration waveforms across body locations (head vs. torso).
- **Key results:** defined the 'sweet spots' of wearable haptic perception across body locations; demonstrated that multi-site dynamic signals improve reaction speed and alert accuracy by 15% over conventional static signals. These perception-threshold and waveform-design methods apply directly to haptic experience design in immersive entertainment venues.

Product Manager & Interaction Design Lead | Smart Construction-Site Safety SaaS, Startup (backed by Allan Myers, a top-20 U.S. contractor) 2023 – Present

- **End-to-end SaaS ownership:** led the full flow from requirements definition (PRD) to MVP delivery; built an LBS geofencing system pairing mobile sensing (Flutter) with cloud decision-making (web dashboard); led a 5-person CS team through cross-platform development and agile iteration.
- **Interaction redesign & trust:** identified the core pain point of visual and auditory signal overload on sites (dense lighting and loud environments) driving alarm desensitization and eroding trust; rebuilt alert thresholds and multimodal interaction logic through field human-factors research, strengthening human-machine trust and markedly improving user acceptance and retention.

Product Designer | Hybrid UAV System, Revolute Robotics (U.S. startup) May 2020 – Aug 2021

- **Structural innovation & engineering trade-offs:** worked with the co-founders as a core contributor to the spherical-exoskeleton structure of a combined Unmanned Aerial Vehicle (UAV) and ground-robot system; led material selection and balanced crash resistance against flight weight.
- **Rapid prototyping:** completed detailed SolidWorks modeling from concept through system-level assembly, accelerating product validation cycles.

Independent Developer & System Architect | Multi-Agent Personal AI System, Personal Project Feb 2026 – Present

- **Project:** built and deployed a personal AI system that helps me capture notes, surface useful connections, curate information, and think through problems; it runs as two Telegram chat assistants plus a web dashboard.
- **How it works:** designed the parts to coordinate by reading and writing shared files rather than calling each other directly, which makes the system easy to watch, debug, and upgrade one piece at a time; each assistant follows rules written in plain-language files, so changes are quick.
- **Human-factors-driven interaction design:** applied human-factors principles directly, turning ideas like managing attention and revealing information gradually into concrete rules for how the assistants behave.
- **Full-stack delivery:** handled the entire build myself (with AI coding tools), including the Python backend, web APIs, chat bots, dashboard, deployment, and cloud operations.
- **Guiding principle:** designed the system to keep the person in control, judging success by how well the person and the system work together rather than by raw model power.

Human Factors Consultant | Cintas Corporation (U.S. publicly listed company) Nov 2023 – May 2024

- **Scope:** ergonomic risk assessment and workstation redesign for production lines in a large industrial laundry plant.
- **Execution:** applied the NIOSH Lifting Equation and RULA/REBA models to quantify musculoskeletal injury risk in high-frequency manual handling.
- **Impact:** designed low-cost fixture solutions that measurably reduced physical-load scores on key processes.

System Architect | Enterprise AI & Digital-Human Platform (B2B)

2026 – Present

- **What it does:** adapted the same shared-file approach for a company setting, so a team's AI tools work together in one shared workspace on a development pipeline.
- **Accountability:** added a running log and an independent review step, so every action and who approved it is recorded and can be checked.

TEACHING EXPERIENCE

- **2019 – 2026:** taught 6+ courses across Virginia Tech and the University of Arizona, reaching 500+ students; roles spanning Teaching Assistant, Substitute Instructor, and Instructor of Record.
- **Instructor of Record, Engineering Economy (VT ISE 2014, 2025):** independently ran the entire course (Canvas, grading rubrics, student communication); student rating 5.35/6 vs. college mean 5.28/6. Also taught Human Factors Research Design & Statistics (substitute instructor, ISE 5624) and Industrial Ergonomics (TA, Spring 2026).

SELECTED PUBLICATIONS

1. **Yang, X.,** & Roofigari-Esfahan, N. (2023). Vibrotactile alerting to prevent accidents in highway construction work zones: An exploratory study. *Sensors*, 23(12), 5651.
2. **Yang, X.,** & Roofigari-Esfahan, N. (2025). Adapting vibrotactile warning systems to individual perception: A human-centered approach to calibration and design. In *Computing in Civil Engineering 2026* (in press).
3. Lim, S., & **Yang, X.** (2023). Real-time vibrotactile feedback system for reducing trunk flexion exposure during construction tasks. *Applied Ergonomics*, 110, 104019.
4. Tajdari, M., **Yang, X.,** Islam, M. S., & Lim, S. (2025). Investigating sensitivity to vibrotactile stimulation in isometric and dynamic tasks: Implications for ergonomic intervention. *Proceedings of the 69th HFES Annual Meeting*. **Best Paper Award.**
5. Tajdari, M., **Yang, X.,** Islam, M. S., & Lim, S. (2026). Investigating sensitivity to vibrotactile stimulation in isometric and dynamic work for effective ergonomic intervention. *International Journal of Industrial Ergonomics*, 115, 103989.

HONORS & AWARDS

- ICTAS Doctoral Research Proposal Award (Roop & Kavita Mahajan Award), a Virginia Tech university-level award recognizing outstanding interdisciplinary doctoral research proposals
- Best Student Research Proposal Award, Human Factors and Ergonomics Society (HFES)
- Best Paper Award, 69th HFES Annual Meeting
- HFES Doctoral Consortium selectee (one of only 2 university-wide)
- 2nd Place (university-wide), '2025 Bad Design Improvement Challenge'